

Report 6: ROMFS Airframe, Parameters, and Flight Configuration

Soaring Parameter Definitions and Default Airframe Setup

PX4 Autosozaring Implementation — Defence Technical Report

Fixed-Wing Autosozaring Project

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1 ROMFS and Parameter Infrastructure

1.1 PX4 Parameter System

PX4 parameters are persistent key-value pairs stored in:

- `/fs/microsd/` (SD card on hardware, or simulated in SITL);
- compiled into the flash image with default values via `PARAM_DEFINE_*` macros in `.c` files.

Each parameter has:

- a unique 16-character name;
- a type (`INT32`, `FLOAT`);
- a default value;
- optional min/max and increment hints for GCS display.

1.2 ROMFS Airframe Files

The `ROMFS/px4fmu_common/init.d/airframes/` directory contains shell scripts that run at boot. They set parameters for a specific vehicle configuration. The airframe file referenced in this project is:

```
1 ROMFS/px4fmu_common/init.d/airframes/4008_gz_advanced_plane
```

This is a Gazebo SITL airframe for a generic advanced plane used during development and simulation.

2 Parameter Definitions in `fw_path_navigation_params.c`

2.1 Original Parameters (Powered-Only)

In the original codebase, the fixed-wing navigation parameters file contained no soaring-specific entries. The nearest relevant parameter was a general airspeed parameter hierarchy:

- `FW_AIRSPD_MIN` – minimum EAS (stall margin) [m/s]
- `FW_AIRSPD_TRIM` – cruise trim EAS [m/s]
- `FW_AIRSPD_MAX` – maximum EAS [m/s]

2.2 Removed Parameter: `FW_GLIDE_MODE`

In the first implementation, a parameter `FW_GLIDE_MODE` was defined as an integer to select between fixed-EAS glide (0) and polar best-glide (1). This was removed in the redesign because the distinction is now captured directly in `soaring_mode`:

Old <code>FW_GLIDE_MODE</code>	New <code>soaring_mode</code>	Meaning
0	1 (<code>SOARING_GLIDE_FIXED</code>)	Fixed EAS
1	2 (<code>SOARING_GLIDE_POLAR</code>)	Polar best-glide

2.3 New Soaring Parameters

2.3.1 FW_GLIDE_AIRSPD

```
1 PARAM_DEFINE_FLOAT(FW_GLIDE_AIRSPD, 10.0f);
```

Unit: m/s (EAS)

Range: [5.0, 30.0]

Purpose: The fixed-EAS glide setpoint for mode 1. Also used as the airspeed reference during thermalling (modes 3 and 4).

The parameter value is an *Equivalent Airspeed (EAS)*, not TAS. TECS performs the conversion:

$$V_{\text{TAS}}^* = k_{\text{EAS2TAS}} \cdot V_{\text{EAS}}^* = \sqrt{\frac{\rho_0}{\rho}} \cdot V_{\text{EAS}}^* \quad (1)$$

Rationale for default 10.0 m/s: A typical mini-UAV has a stall speed around 8–9 m/s EAS. A 10 m/s glide speed provides a safe margin above stall while keeping the glide angle shallow (small $\gamma = -w_s/V$).

2.3.2 FW_POLAR_A

```
1 PARAM_DEFINE_FLOAT(FW_POLAR_A, 0.003f);
```

Unit: s/m

Description: Quadratic coefficient a of the glide polar:

$$w_s(V) = aV^2 + b \quad (2)$$

Physics: a scales with induced drag. Its value is:

$$a = \frac{k}{\pi e AR \frac{1}{2} \rho S V^2 / (mg)} = \frac{2(mg)k}{\pi e AR \rho S} \quad (3)$$

(approximate form). For a 2 kg, 0.8 m² wing, span 2 m: $a \approx 0.003$.

2.3.3 FW_POLAR_B

```
1 PARAM_DEFINE_FLOAT(FW_POLAR_B, 0.5f);
```

Unit: m/s

Description: Constant term b of the glide polar (minimum sink rate).

Physics: b corresponds to parasitic drag at very low speed:

$$b \approx \frac{W}{T_{\text{max}}} w_{s,0} \quad (4)$$

For a 2 kg aircraft with $V_{\text{min-sink}} \approx 0.5$ m/s: $b = 0.5$.

2.3.4 Best-Glide Derivation from A and B

$$V_{\text{EAS,polar}}^* = \sqrt{\frac{b}{a}} = \sqrt{\frac{0.5}{0.003}} \approx 12.9 \text{ m/s} \quad (5)$$

The glide ratio at this speed:

$$\left. \frac{L}{D} \right|_{\text{max}} = \frac{V^*}{w_s(V^*)} = \frac{\sqrt{b/a}}{a(b/a) + b} = \frac{1}{2\sqrt{ab}} = \frac{1}{2\sqrt{0.003 \times 0.5}} = \frac{1}{0.0775} \approx 12.9 \quad (6)$$

2.3.5 FW_GLIDE_I_DECAY

```
1 PARAM_DEFINE_FLOAT(FW_GLIDE_I_DECAY, 10.0f);
```

Unit: s

Description: Throttle integrator decay time constant τ during gliding:

$$I_t(t) = I_t(0) e^{-t/\tau} \quad (7)$$

After $3\tau = 30$ s, the integrator retains only $e^{-3} \approx 5\%$ of its pre-glide value.

2.3.6 FW_GLIDE_RAMP_T

```
1 PARAM_DEFINE_FLOAT(FW_GLIDE_RAMP_T, 3.0f);
```

Unit: s

Description: Duration of the symmetric entry/exit throttle ramps.

Sizing: A 3 s ramp at 50 Hz corresponds to 150 control cycles. With $\Delta\delta_t = 0.4$ (typical cruise throttle), the rate is 0.133 per second.

2.3.7 FW_THERMAL_BANK

```
1 PARAM_DEFINE_FLOAT(FW_THERMAL_BANK, 40.0f);
```

Unit: degrees

Description: Default bank angle for thermalling when the companion does not specify `bank_angle_cmd`.

Physics: At $\phi = 40^\circ$, the load factor and turn radius are:

$$n_z = 1/\cos(40^\circ) = 1.305 \quad (8)$$

$$R_{\min} = \frac{V^2}{g \tan \phi} = \frac{10^2}{9.81 \times \tan(40^\circ)} \approx 12.2 \text{ m} \quad (9)$$

2.3.8 FW_ALT_MIN and FW_ALT_MAX

```
1 PARAM_DEFINE_FLOAT(FW_ALT_MIN, 50.0f);  
2 PARAM_DEFINE_FLOAT(FW_ALT_MAX, 200.0f);  
3 PARAM_DEFINE_FLOAT(FW_ALT_HYST, 20.0f);
```

Safety altitude bounds:

- FW_ALT_MIN: soaring inhibited below this AGL altitude.
- FW_ALT_MAX: thermal mode forced off above this altitude.
- FW_ALT_HYST: hysteresis on the ceiling to prevent chatter.

3 ROMFS Airframe: 4008_gz_advanced_plane

3.1 Purpose

The ROMFS airframe script sets default parameter values for the Gazebo SITL “advanced plane” model. Changes were made to add soaring defaults.

3.2 Added Parameter Defaults

```

1 # Autos soaring parameters
2 param set FW_GLIDE_AIRSPD 10.0
3 param set FW_POLAR_A 0.003
4 param set FW_POLAR_B 0.5
5 param set FW_GLIDE_I_DECAY 10.0
6 param set FW_GLIDE_RAMP_T 3.0
7 param set FW_THERMAL_BANK 40.0
8 param set FW_ALT_MIN 50.0
9 param set FW_ALT_MAX 200.0
10 param set FW_ALT_HYST 20.0

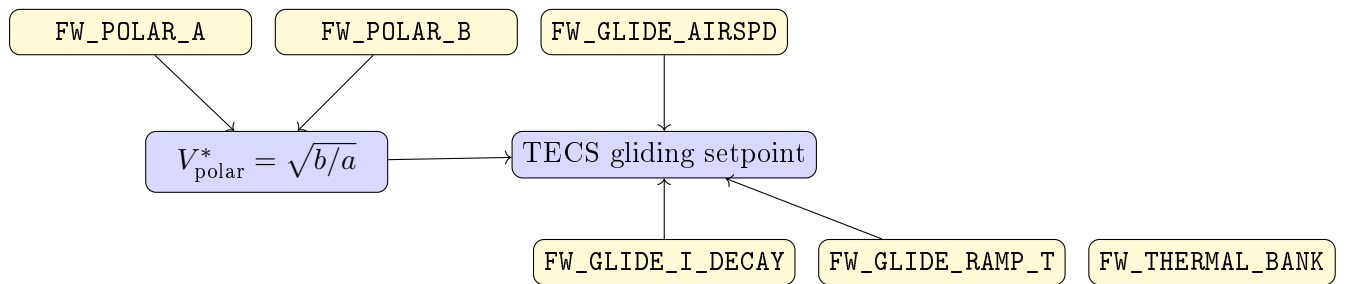
```

3.3 Justification for Default Values

Parameter	Value	Unit	Rationale
FW_GLIDE_AIRSPD	10.0	m/s EAS	~ 25% above stall
FW_POLAR_A	0.003	s/m	Typical mini-UAV induced drag
FW_POLAR_B	0.5	m/s	Typical mini-UAV parasitic drag
FW_GLIDE_I_DECAY	10	s	5% residual after 30 s
FW_GLIDE_RAMP_T	3	s	150 cycles at 50 Hz
FW_THERMAL_BANK	40	deg	$R_{\min} = 12.2$ m at 10 m/s
FW_ALT_MIN	50	m AGL	Terrain clearance
FW_ALT_MAX	200	m AGL	Typical soaring ceiling
FW_ALT_HYST	20	m	~ 10% of ceiling range

4 Complete Parameter Dependency Graph

The following shows how each parameter flows into the control pipeline:



5 Summary

Aspect	Original	Modified
Soaring parameters	None	9 new parameters
FW_GLIDE_MODE	Defined (integer, polar select)	Removed (mode in message)
Airframe defaults	No soaring values	Added to 4008_gz_advanced_plane
Polar model	Not present	$w_s = aV^2 + b$ with a, b params